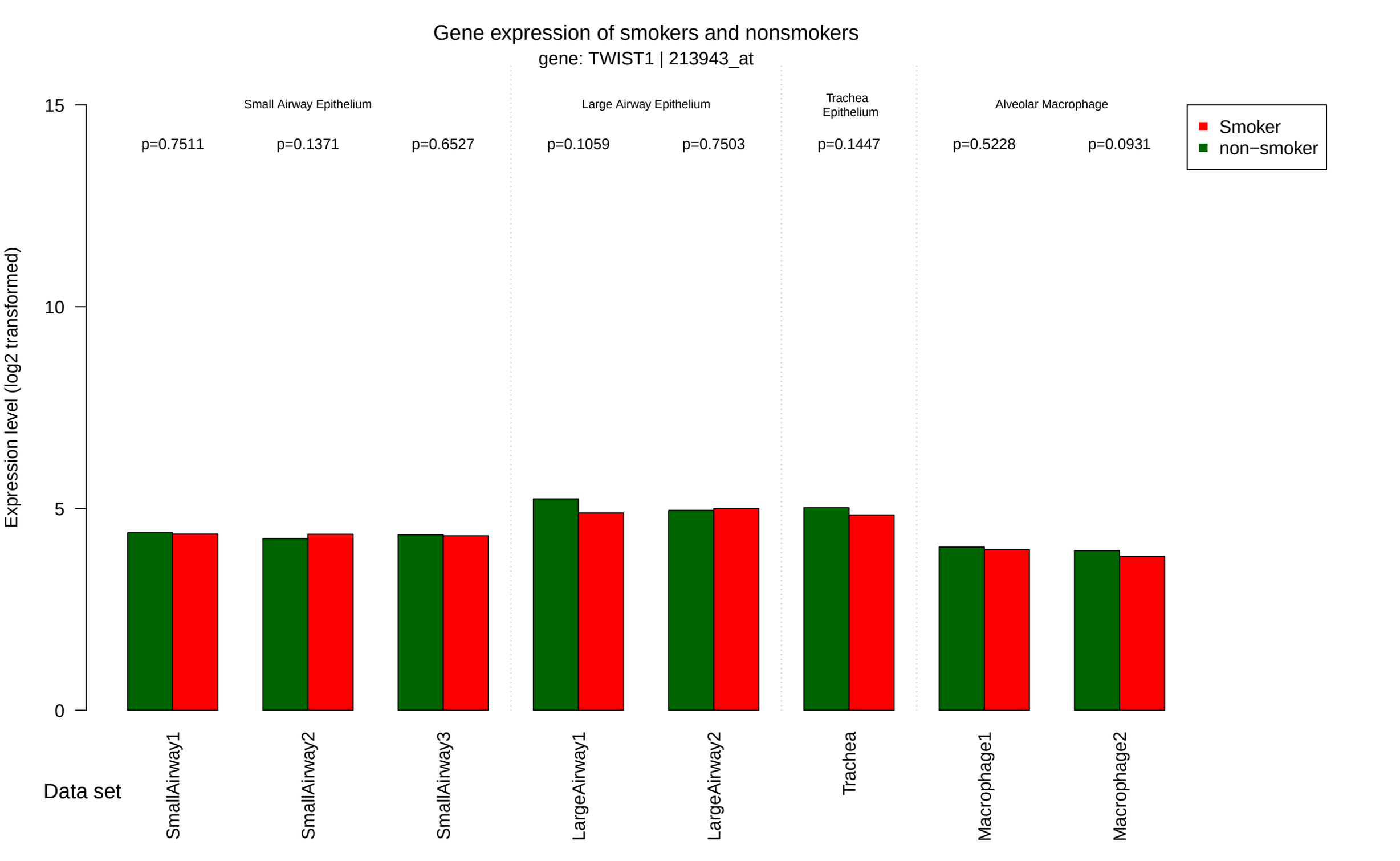
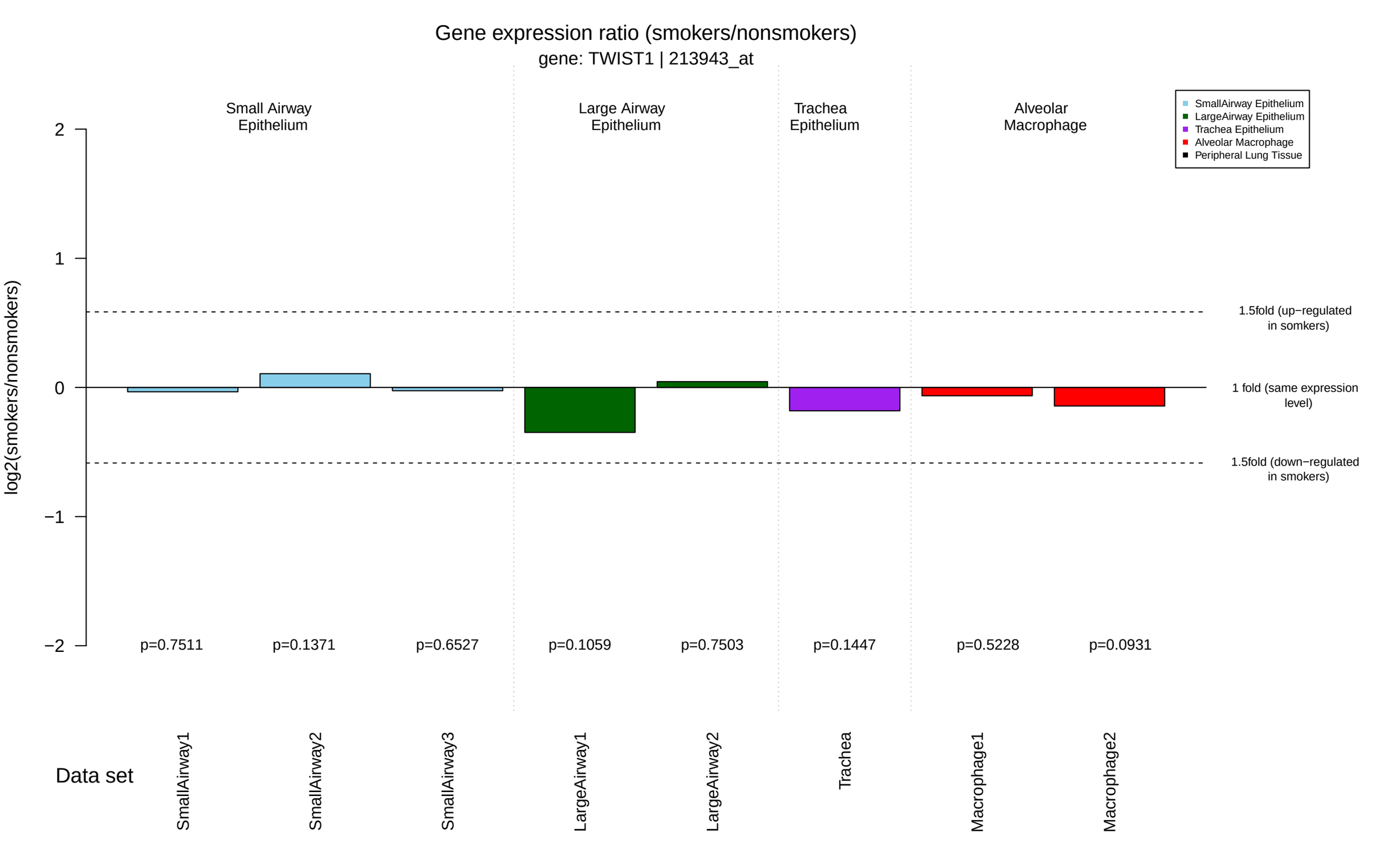
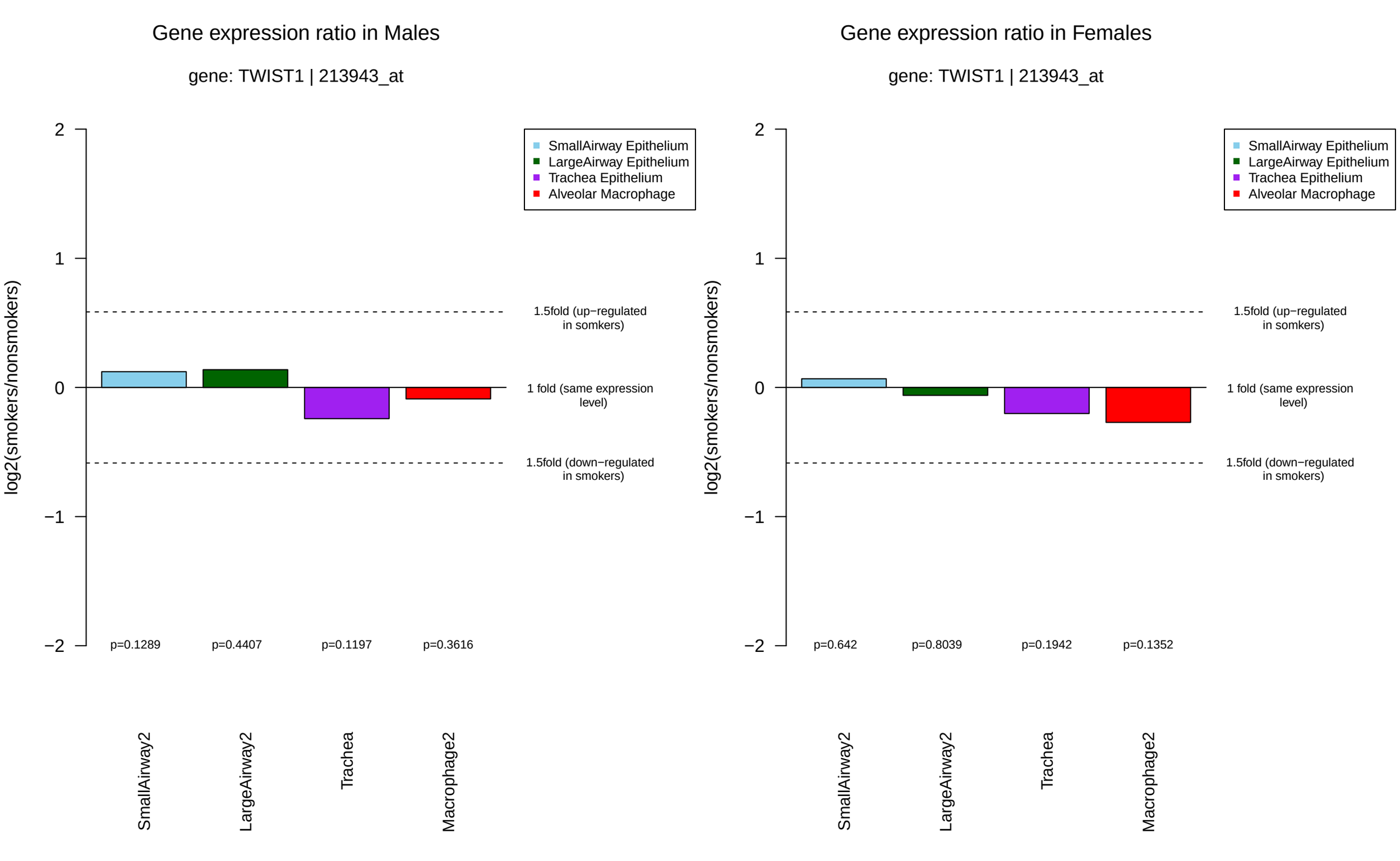
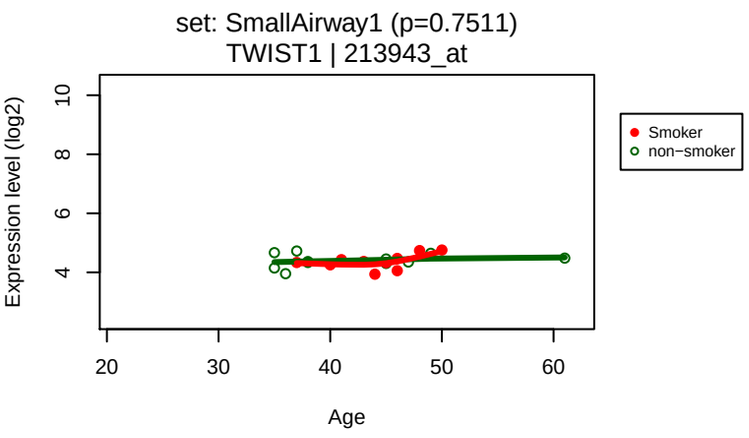
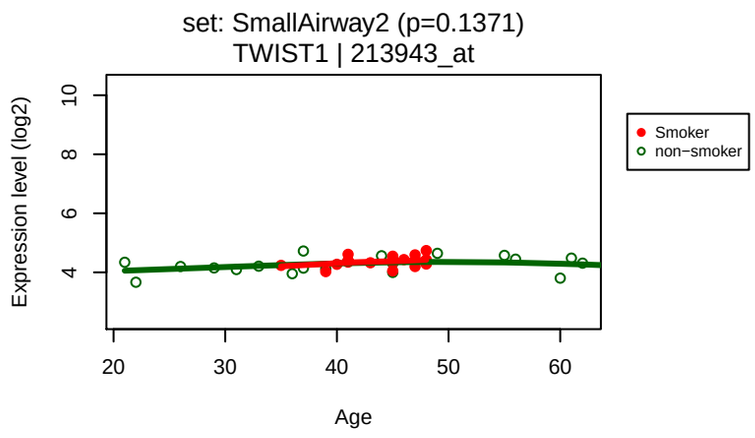
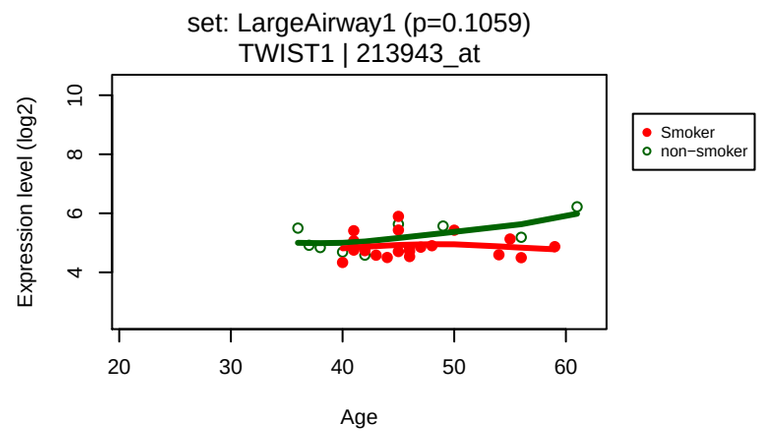

Gene expression of smokers and nonsmokers
gene: TWIST1 | 213943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7511
p=0.1371
p=0.6527
p=0.1059
p=0.7503
p=0.1447
p=0.5228
p=0.0931
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

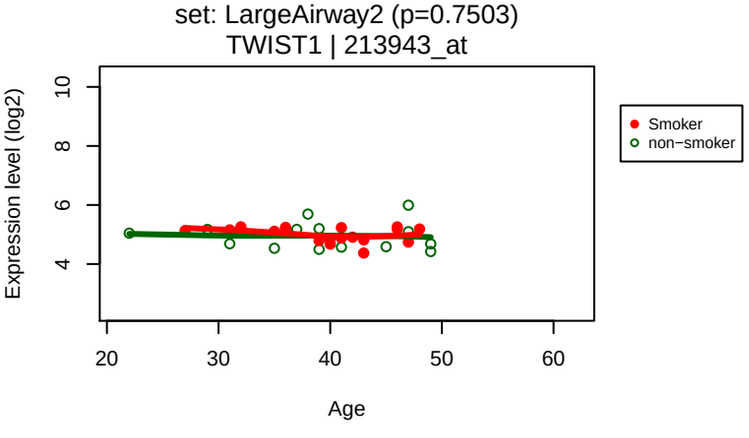

Gene expression ratio (smokers/nonsmokers)
gene: TWIST1 | 213943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7511
p=0.1371
p=0.6527
p=0.1059
p=0.7503
p=0.1447
p=0.5228
p=0.0931
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

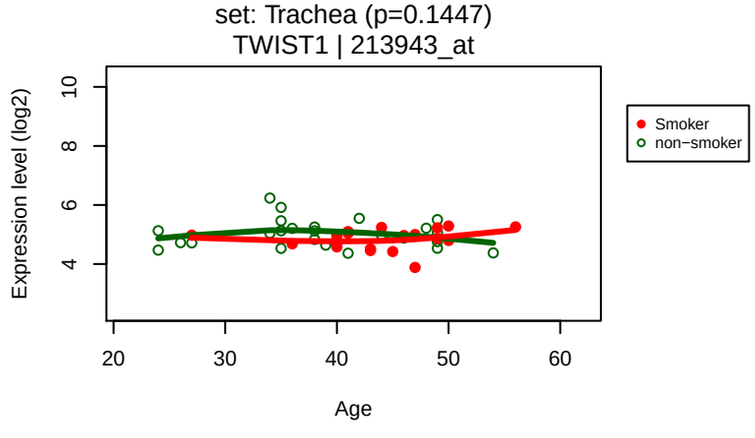

Gene expression ratio in Males
gene: TWIST1 | 213943_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1289
p=0.4407
p=0.1197
p=0.3616
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TWIST1 | 213943_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.642
p=0.8039
p=0.1942
p=0.1352
SmallAirway2
LargeAirway2
Trachea
Macrophage2

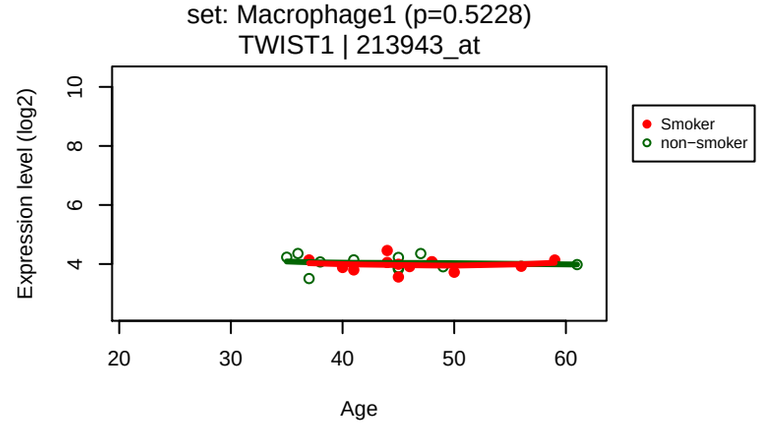

set: SmallAirway1 (p=0.7511)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker

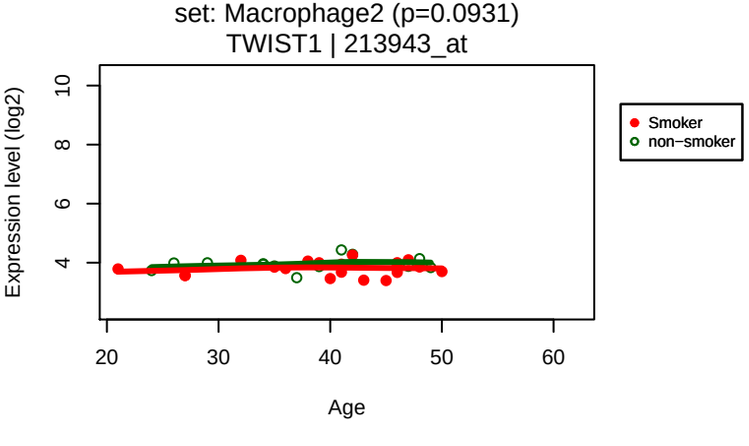

set: SmallAirway2 (p=0.1371)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker

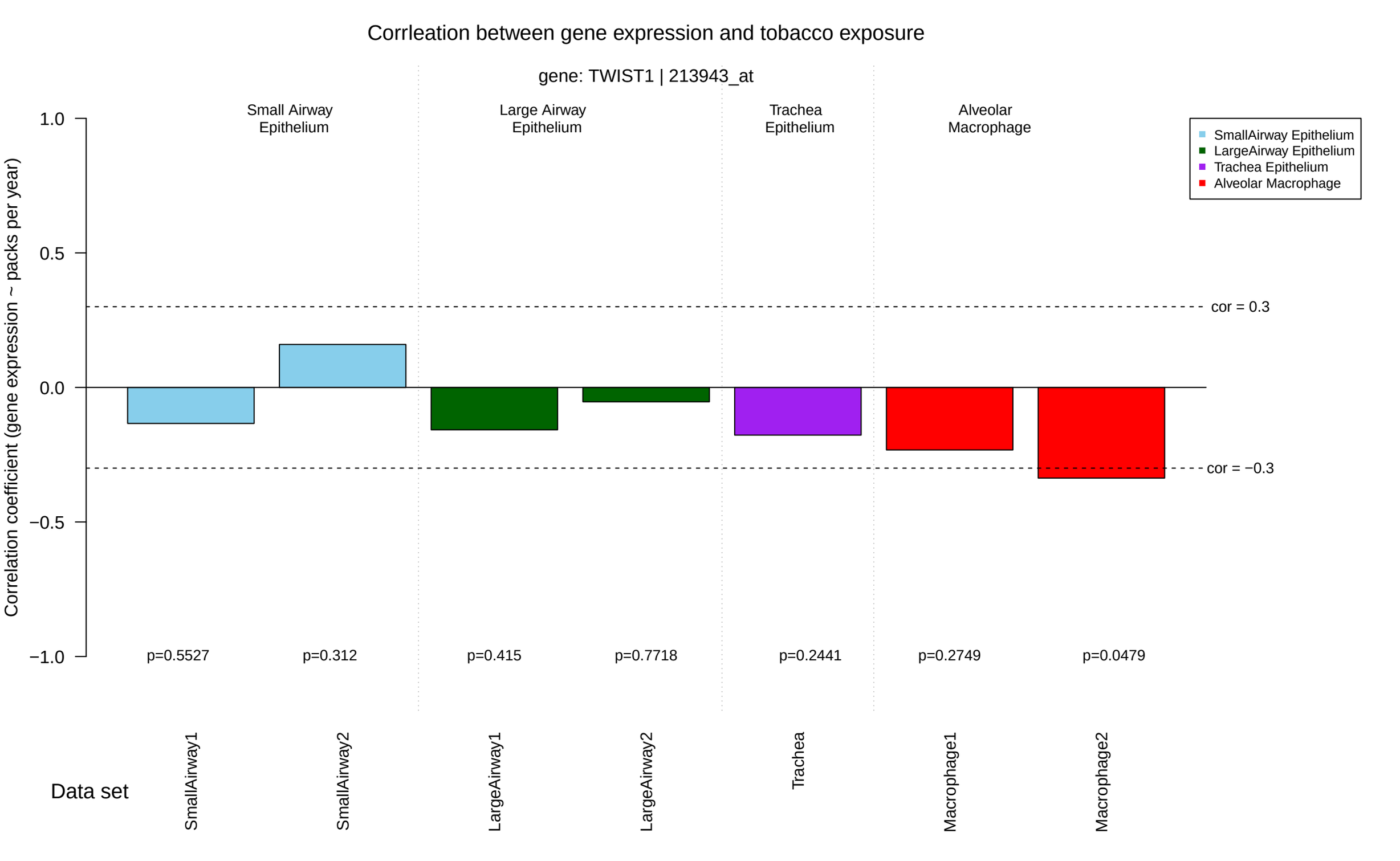

set: LargeAirway1 (p=0.1059)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7503)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1447)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5228)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0931)
TWIST1 | 213943_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TWIST1 | 213943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5527
p=0.312
p=0.415
p=0.7718
p=0.2441
p=0.2749
p=0.0479
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2